MARKETMIND AI: Project Documentation

1. Introduction

In today's stock market ecosystem, a huge number of retail investors are actively participating. However, most of them struggle to interpret financial data such as company reports, stock charts, and market news. While platforms like ET Markets provide a large amount of information, the challenge lies in converting that information into meaningful decisions.

Our project, **MarketMind AI**, focuses on solving this exact problem. The idea is simple : instead of showing users raw data, we provide **clear, understandable, and actionable insights**.

2. Objective

The main objective of this project is to build an AI-based system that can:

- Analyze stock market data from multiple perspectives
- Generate clear investment suggestions (Buy / Hold / Sell)
- Explain the reasoning in simple language
- Present insights in both text and video formats

3. Overall Approach

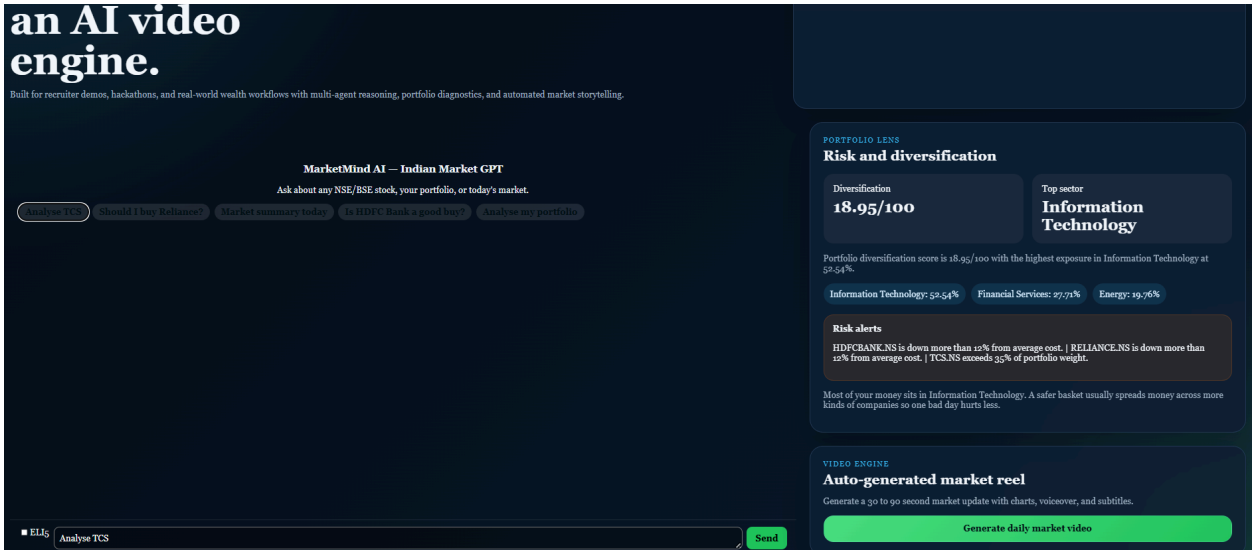
The system is designed in a modular way, where different components handle different aspects of analysis. Instead of relying on a single model, we use multiple "agents" that specialize in specific types of data.

All these components are then connected through a central pipeline that processes user queries and generates the final output.

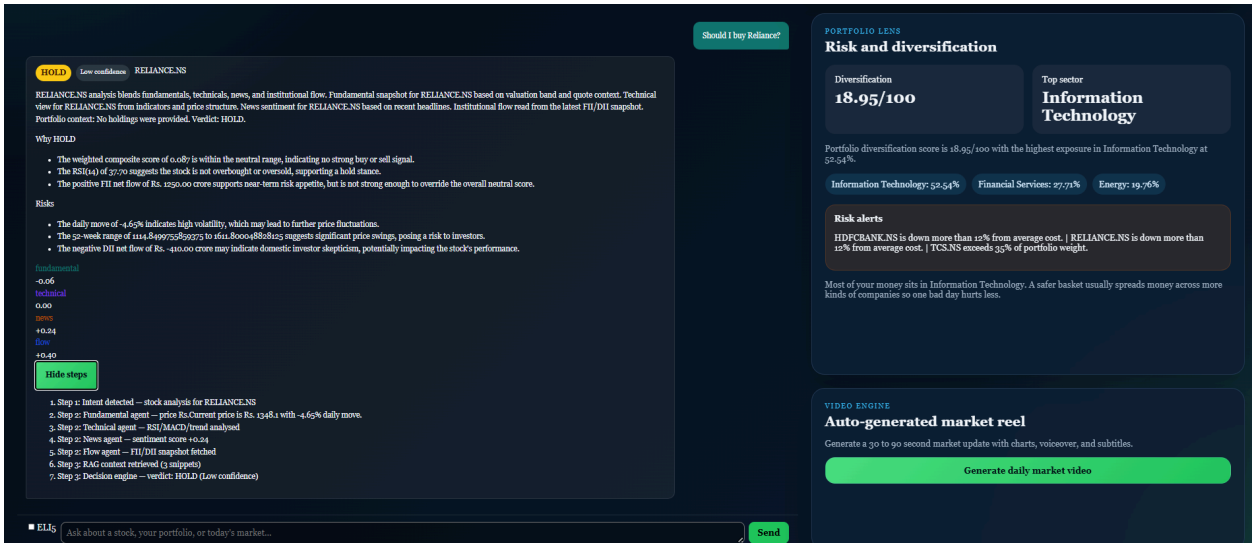
4. System Architecture

The workflow of the system is as follows:

1. The user asks a query (for example: "Should I buy TCS?")
2. The system identifies the intent and extracts the stock symbol
3. Multiple agents analyze the stock from different perspectives
4. The decision engine combines all signals
5. The result is explained using an LLM
6. Output is shown in text and can also be converted into a video



Should I buy Reliance?.....



5. Core Components

5.1 Multi-Agent System

The system uses four main agents:

- **Fundamental Agent**
  Looks at company performance such as revenue, profit, and financial strength. This helps in understanding long-term potential.
- **Technical Agent**
  Uses indicators like RSI and trends to understand short-term price movement.
- **News Agent**
  Analyzes recent news and headlines to determine market sentiment.
- **Flow Agent**
  Tracks institutional activity such as FII and DII investments to understand market confidence.

Each agent produces a score and key observations.

5.2 Orchestrator

The orchestrator acts as a controller. It calls all agents, collects their outputs, and prepares the data for the next stage.

5.3 Decision Engine

This is the most important part of the system.

Instead of simply averaging results, we implemented a **weighted scoring system**, where:

- Fundamentals are given higher importance
- Technicals capture short-term trends
- News and flows provide supporting signals

The engine also handles cases where signals conflict. For example, if technical indicators are positive but fundamentals are weak, the system avoids giving overly optimistic recommendations.

The final output includes:

- Verdict (Buy / Hold / Sell)
- Confidence level
- Key reasons
- Possible risks

5.4 LLM Integration

A language model is used to convert technical outputs into natural language explanations. It helps in:

- Making responses easy to understand
- Providing simplified explanations (ELI5 mode)
- Generating scripts for videos

5.5 AI Video Engine

One of the key features of our project is the ability to convert market insights into short videos.

The process works as follows:

1. Market data is collected (indices, movers, news, flows)
2. A structured script is generated using AI
3. The script is converted into speech using text-to-speech
4. The video builder creates multiple scenes from the script
5. The final output is a short, reel-style video

This makes the platform more engaging and accessible, especially for users who prefer visual content.

6. Tech Stack

- **Backend:** Python, FastAPI
- **Frontend:** React / Next.js
- **AI/ML:** LLM APIs, custom decision logic
- **Data Sources:** Market APIs, news APIs
- **Video Processing:** MoviePy, gTTS

7. Example Workflow

User Input:
"Should I buy TCS?"

System Output:

- Verdict: HOLD
- Confidence: Medium
- Reason: Mixed signals with weak fundamentals
- Suggestion: Wait before making a decision

The same information can also be presented as a short video summarizing the market condition.

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8. Key Highlights

- Uses multiple specialized agents instead of a single model
- Provides explainable and transparent reasoning
- Focuses on actionable insights rather than raw data
- Includes a video generation feature for better engagement

9. Future Scope

- Personalized recommendations based on user portfolio
- More advanced predictive models
- Mobile application integration
- Improved video quality with charts and animations

10. Conclusion

MarketMind AI aims to simplify stock market analysis for retail investors by combining data analysis, AI reasoning, and content generation.

Instead of overwhelming users with data, the system focuses on answering a simple question:

> “What should I do next?”